12TH CHEMISTRY · CHAPTER 18 · FULL CHAPTER NOTES

Transition Metals

Topic-wise short and long questions with answers from the complete chapter

INTRODUCTION

Q.1 [Short] What are transition metals? How are they classified into d-block and f-block?

Transition metals are found in the d and f blocks of the modern periodic table. They are characterised by high density, conductivity and the ability to form colourful compounds. If their outermost electrons are being filled in d-orbitals they are d-block elements; if in f-orbitals, f-block elements. The first row (3d series) is the most common in daily life.

Q.2 [Short] Write the importance of iron, copper, silver and gold.

Iron is the backbone of human civilization: it played a key role in the industrial revolution for durable tools and modern steel infrastructure. Copper, silver and gold have been symbols of wealth and prestige since ancient ages, used for coinage and jewellery. Today they remain essential in high-end electronics and dentistry due to their exceptional corrosion resistance and conductivity.

18.1 FIRST ROW TRANSITION ELEMENTS (3d)

Q.3 [Short] Define transition elements. Why are they so called?

First row transition elements are the elements found in the d-block of the Periodic Table that have **partially filled d-orbitals in any of their stable oxidation states**. They are called transition elements because they show transitions between variable oxidation states.

Q.4 [Short] Why are zinc and scandium not classified as transition elements?

The d-block is composed of groups 3 to 12, but zinc and scandium cannot be classified as transition elements. Zinc forms only one ion, Zn^{2+}, which has a **complete** 3d subshell: [Ar] $3d^{10}$ $4s^{0}$. Scandium forms only Sc^{3+}, [Ar] $3d^{0}$ $4s^{0}$, whose d-subshell is **empty**. Neither has a stable ion with a partially filled d-subshell.

18.1.1 Electronic Configuration of Transition Metals and their Ions

Q.5 [Long] Write the electronic configurations of the first row transition elements (Ti to Cu) and of their M^{2+} and M^{3+} ions.

Element	Symbol	Atomic No.	M^{0}	M^{2+}	M^{3+}
Titanium	Ti	22	[Ar] $3d^{2}$ $4s^{2}$	[Ar] $3d^{2}$ $4s^{0}$	[Ar] $3d^{1}$ $4s^{0}$
Vanadium	V	23	[Ar] $3d^{3}$ $4s^{2}$	[Ar] $3d^{3}$ $4s^{0}$	[Ar] $3d^{2}$ $4s^{0}$
Chromium	Cr	24	[Ar] $3d^{5}$ $4s^{1}$	[Ar] $3d^{4}$ $4s^{0}$	[Ar] $3d^{3}$ $4s^{0}$

Element	Symbol	Atomic No.	M^0	M^{2+}	M^{3+}
Manganese	Mn	25	[Ar] $3d^5\ 4s^2$	[Ar] $3d^5\ 4s^0$	[Ar] $3d^4\ 4s^0$
Iron	Fe	26	[Ar] $3d^6\ 4s^2$	[Ar] $3d^6\ 4s^0$	[Ar] $3d^5\ 4s^0$
Cobalt	Co	27	[Ar] $3d^7\ 4s^2$	[Ar] $3d^7\ 4s^0$	[Ar] $3d^6\ 4s^0$
Nickel	Ni	28	[Ar] $3d^8\ 4s^2$	[Ar] $3d^8\ 4s^0$	[Ar] $3d^7\ 4s^0$
Copper	Cu	29	[Ar] $3d^{10}\ 4s^1$	[Ar] $3d^9\ 4s^0$	[Ar] $3d^8\ 4s^0$

Q.6 [Short] **Why do chromium and copper show anomalous electronic configurations?**

The electronic configuration of the transition elements usually follows the Aufbau principle, except in **chromium** and **copper**. These two show an anomaly because they gain extra stability from a half-filled (Cr: $3d^5\ 4s^1$) and a completely filled (Cu: $3d^{10}\ 4s^1$) d-subshell. This makes the electron cloud more symmetrical and gives a better shielding effect to the nucleus.

Q.7 [Short] **Which electrons are lost first when a 3d metal forms an ion? Give an example.**

The 4s electrons are lost first, then the 3d electrons. For example, Fe is [Ar] $3d^6\ 4s^2$; Fe^{2+} is [Ar] $3d^6\ 4s^0$ and Fe^{3+} is [Ar] $3d^5\ 4s^0$.

18.2 PROPERTIES OF TRANSITION ELEMENTS

18.2.1 Physical Properties

Q.8 [Long] **Describe the general physical properties of the first row transition elements.**

- They exist as hard and rigid solids.
- Except copper, which is reddish brown, the others occur as silvery or grey lustrous metals.
- Their atomic radius usually decreases from left to right; at the end it increases a little. This increase may be associated with the stability acquired by copper due to its completely filled d-subshell.
- They have high melting and boiling points, which show a variable trend from left to right.
- They are good conductors of electricity and heat.
- They have high densities that generally increase across the row from titanium to copper, due to a regular build-up of nuclear charge and shrinkage of atomic size.
- They possess a shiny surface and are malleable (can be hammered into thin sheets) and ductile (can be drawn into wires).

- Most of them and their compounds are paramagnetic due to the presence of unpaired d electrons.

Data read from Figure 18.1 of the textbook:

	Ti	V	Cr	Mn	Fe	Co	Ni	Cu
Atomic radius (pm)	147	134	128	127	126	125	124	128
Ionic radius (pm)	86	79	80	81	76	74	69	73
Melting point (°C)	1688	1910	1907	1246	1538	1495	1455	1085
Boiling point (°C)	3287	3407	2672	2061	2862	2870	2913	2562

Q.9 [Short] Why are most transition metals and their compounds paramagnetic?

Because they contain unpaired d electrons. Substances with unpaired electrons are attracted by a magnetic field.

Q.10 [Short] Why does the density of 3d metals increase from titanium to copper?

Across the row there is a regular build-up of nuclear charge (and atomic mass) while the atomic size shrinks. More mass in a smaller volume gives a higher density.

Q.11 [Short] Why does the atomic radius increase a little at the end of the 3d series?

The atomic radius decreases from left to right, but at the end (copper) it increases a little. This may be associated with the stability acquired by copper due to its completely filled d-subshell.

18.2.2 Chemical Properties

Q.12 [Short] State four characteristic chemical properties of transition elements.

- They have **variable oxidation states**, as they can lose varying numbers of electrons to form stable ions (e.g., Fe^{2+} and Fe^{3+}).
- They have a high tendency to form **coordination compounds (complexes)** by accepting lone pairs of electrons from ligands into vacant d-orbitals.
- They form **coloured ions and complexes** due to the transition of electrons from a lower to a higher energy level as a result of absorption of light.
- The metals and their compounds behave as **catalysts** because they can easily change oxidation states.

studysapiens.com

18.2.3 Variable Oxidation States

Q.13 [Long] **Why do transition elements show variable oxidation states? Give the common oxidation states of the 3d elements.**

First row transition elements show variable oxidation states because the **3d and 4s subshells have comparable (very similar) energy**. The energy difference between the inner 3d subshell and the outermost 4s subshell is very small. So, like the 4s electrons, the 3d electrons can also easily be lost: both take part in chemical bonding. A 3d metal atom can lose:

- only the 4s electrons, usually giving the common **+2** oxidation state;
- the 4s electrons plus a variable number of 3d electrons, giving the **higher** oxidation states.

Among these metals Mn has the maximum oxidation state of +7. The oxidation state varies in a parabolic trend from left to right within the d-block.

Common oxidation states (Table 18.2; the highest of each element is in bold):

Ti	V	Cr	Mn	Fe	Co	Ni	Cu
+2, +3, **+4**	+2, +3, +4, **+5**	+2, +3, +4, +5, **+6**	+1 to +6, **+7**	+1 to +5, **+6**	+2, +3, +4, **+5**	+2, +3, **+4**	+1, +2, **+3**

Q.14 [Short] **Which 3d element shows the highest oxidation state, and why?**

Manganese, +7. Mn is [Ar] $3d^5$ $4s^2$; it can use all its two 4s and five 3d electrons in bonding.

Quick Check 18.1

Q.15 [Short] **a) Write down electronic configuration of Co^{2+} and Co^{3+}, and find the number of unpaired electrons in d-orbitals.**

Co^{2+}: [Ar] $3d^7$ $4s^0$, with **3** unpaired d electrons. Co^{3+}: [Ar] $3d^6$ $4s^0$, with **4** unpaired d electrons (the five d-orbitals are first filled singly, then pairing starts).

Q.16 [Short] **b) Why are the density and melting points of transition metals much higher than Group 1 metals?**

Transition metal atoms are smaller and heavier, with a higher nuclear charge, so more mass is packed into a small volume (high density). Both their 4s and 3d electrons take part in metallic bonding, while a Group 1 metal gives only one electron. The metallic bonds are therefore much stronger and need more heat to break (high melting point).

Q.17 [Short] **c) The Sc^{3+} ion is [Ar] $3d^0$ $4s^0$. Why is scandium not classified as a transition element?**

A transition element must have a partially filled d-subshell in a stable oxidation state. Scandium forms only Sc^{3+}, whose 3d subshell is empty ($3d^0$), so it is not classified as a transition element.

studysapiens.com

18.2.4 Transition Metals as Catalysts

Q.18 [Long] **Explain why transition elements behave as catalysts. Give examples of homogeneous and heterogeneous catalysis.**

1. Variable oxidation states (usually homogeneous catalysis): Transition elements can easily switch between their stable oxidation states (e.g., $Fe^{2+} \rightleftharpoons Fe^{3+}$). The metal ion forms an unstable intermediate compound with a reactant molecule. The intermediate quickly reacts to form the final product and regenerates the catalyst in its original oxidation state, providing a pathway of lower activation energy. Example: nickel salts used in the preparation of acrylic acid from acetylene involve the oxidation states Ni(0) and Ni(II).

2. Vacant, energetically accessible d-orbitals (generally heterogeneous catalysis): Reactant molecules attach to the catalyst surface and form a temporary activated complex, which holds the reactants in the correct orientation and lowers the activation energy. After the reaction the products desorb and the catalyst is ready for the next cycle. Example: solid iron accelerates the reaction between nitrogen and hydrogen to form ammonia in the Haber process.

Q.19 [Short] **Differentiate between homogeneous and heterogeneous catalysis by transition metals.**

Homogeneous catalysis	Heterogeneous catalysis
Depends on the ability to switch between stable oxidation states.	Depends on vacant, energetically accessible d-orbitals on the metal surface.
The metal ion forms an unstable intermediate compound with the reactant and is then regenerated.	Reactants attach to the surface, form a temporary activated complex, react, and the products desorb.
Example: nickel salts, Ni(0)/Ni(II), in making acrylic acid from acetylene.	Example: solid iron in the Haber process.

Quick Check 18.2

Q.20 [Short] **a) How Vanadium can show the range of oxidation states from +2 to +5?**

Vanadium is [Ar] $3d^3\ 4s^2$. Its 3d and 4s subshells have very similar energy. Losing the two 4s electrons gives +2; losing in addition one, two or three 3d electrons gives +3, +4 and +5.

Q.21 [Short] **b) What is the role of a ligand in forming a complex ion, and what type of bond is formed?**

The ligand acts as a Lewis base: it donates a lone pair of electrons into a vacant d-orbital of the central metal atom or ion. The bond formed is a coordinate covalent (dative) bond.

Q.22 [Short] c) Why transition metals use both 4s and 3d electrons in bonding, unlike s-block elements.

In transition metals the energy difference between the inner 3d subshell and the outer 4s subshell is very small, so the 3d electrons can be lost as easily as the 4s electrons. In s-block elements the electrons below the outer s-subshell belong to a stable noble-gas core of much lower energy, so only the s electrons are used.

Q.23 [Short] d) Write the electronic configuration for Fe^{3+}. Why is the Fe^{2+}/Fe^{3+} change useful in catalysis?

Fe^{3+}: [Ar] $3d^5$ $4s^0$. Iron switches easily between Fe^{2+} and Fe^{3+}. So the ion can form an unstable intermediate with a reactant and then be regenerated in its original oxidation state, giving a reaction pathway of lower activation energy.

Q.24 [Short] e) Which properties of transition metals enable them to be used as heterogeneous catalysts and homogeneous catalysts.

Heterogeneous catalysts: vacant, energetically accessible d-orbitals, which let reactant molecules attach to the metal surface (e.g., iron in the Haber process). **Homogeneous catalysts:** more than one stable oxidation state, between which the metal ion can easily switch (e.g., Ni(0)/Ni(II)).

18.3 TRANSITION METAL COMPLEXES

Q.25 [Short] Why do transition elements form complex ions?

Transition elements form complex ions because they have **vacant (empty) d-orbitals that are energetically accessible** to accept electron pairs. The central metal ion (e.g., Fe^{3+} or Cu^{2+}) has partially filled or empty (n–1)d orbitals whose energy is low enough to take part in bond formation. These orbitals accept the lone pairs donated by the ligands, forming coordinate covalent (dative) bonds.

$$Cu^{2+}_{(aq)} + 4NH_{3(aq)} \rightarrow [Cu(NH_3)_4]^{2+}$$

Q.26 [Short] Define a metal complex (coordination compound).

A **metal complex** (coordination compound) is a molecule or ion that forms when a central metal atom or ion is surrounded by a number of electron donor species (molecules or ions). The bond between the metal and each ligand is a **coordinate covalent (dative) bond**: the ligand donates the electron pair (Lewis base) and the metal accepts it (Lewis acid).

Q.27 [Short] What is a ligand? Why does it act as a Lewis base?

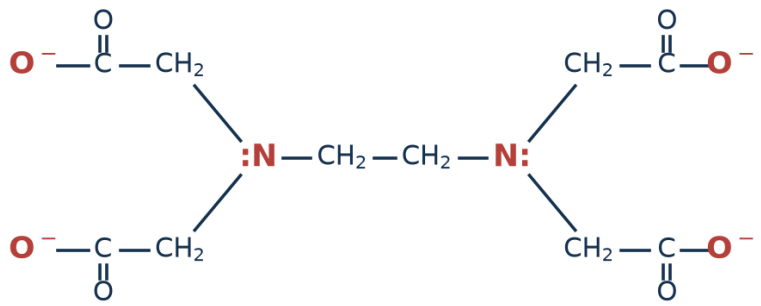
A **ligand** is a molecule or ion (e.g., H_2O, NH_3 or Cl^-) that possesses at least one lone pair of electrons and forms a coordinate covalent (dative) bond to the central metal atom or ion in a complex. The word comes from the Latin “ligare”, meaning “to bind”. It acts as a Lewis base.

Q.28 [Long] Classify ligands on the basis of their charge.

Type	Description	Examples and naming
Anionic	Negatively charged electron pair donors; the most abundant type	Cl^-, CN^-, OH^-, $C_2O_4^{2-}$; names usually end in “-o” (chloro, cyano)
Neutral	Neutral molecules that donate an electron pair	H_2O (aqua), NH_3 (ammine), CO (carbonyl); names usually remain the same
Cationic	Positively charged electron pair donors; rare	NO^+ (nitrosonium), $NH_2–NH_3^+$ (hydrazinium); names often end in “-ium”

Q.29 [Long] What are monodentate, bidentate and polydentate ligands? Give examples.

Type	Electron pairs donated	Examples
Monodentate	One; a single atom of the ligand forms the dative bond	H_2O, NH_3, CO, Cl^-, OH^-, CN^-
Bidentate	Two	1,2-diaminoethane (ethylenediamine, $H_2NCH_2CH_2NH_2$, *en*): each N donates a pair; ethanedioate (oxalate, $C_2O_4^{2-}$): two O atoms, one from each carboxyl group
Polydentate	More than two dative bonds to a single metal	Ethylenediaminetetraacetate ion (EDTA): hexadentate, six donor atoms (four oxygen and two nitrogen)



Electron pair donating sites (red): 2 nitrogen atoms + 4 oxygen atoms = 6

EDTA, a hexadentate ligand

Q.30 [Short] Define coordination number with an example.

The **coordination number** is the total number of dative bonds formed between the ligands and the central metal atom or ion in a complex. It determines the overall structure of the complex. Example: in $[Fe(CN)_6]^{4-}$ the Fe^{2+} ion is bonded to six CN^- ligands, so its coordination number is 6.

Q.31 [Short] What is a coordination sphere? How is its charge found?

A **coordination sphere** is the central metal atom or ion and the ligands directly bonded to it. It defines the immediate chemical environment and is usually written inside square brackets. The charge on the coordination sphere is the arithmetic sum of the charges of the central metal and the ligands; the charge on the metal is the difference between the charge on the sphere and the total charge of the ligands.

In $[Fe(CN)_6]^{4-}$: central metal ion = Fe^{2+}; ligand = CN^-; coordination number = 6; total charge of ligands = –6; charge on metal = +2; charge on coordination sphere = –4.

In this paragraph the textbook prints the ion as $[Fe(CN)_6]^{4+}$; Figure 18.2 and Table 18.3 give the correct charge, 4–.

18.4 PREDICTING FORMULA AND CHARGE OF A COMPLEX ION

Q.32 [Long] How are the formula and charge of a complex ion predicted? Explain with cationic, anionic and neutral examples.

The formula and charge of a complex ion are found by combining the central metal ion, its oxidation state and the ligands with their charges. The general structure is $\mathbf{[M(L)_x]^{\pm y}}$, where M is the central metal ion, L the ligand, x the number of ligands (fixed by the coordination number) and ±y the overall charge.

1. Write the metal first, then the ligand with its number as a subscript, all inside square brackets.
2. Charge of complex = metal charge + (number of ligands × ligand charge).

Complex type	Cationic	Anionic	Neutral
Metal ion and oxidation state	Co, +3	Fe, +2	Ni, 0
Ligand (type, charge)	NH_3, ammine (neutral, 0)	CN^-, cyano (anionic, –1)	CO, carbonyl (neutral, 0)
Coordination number	6	6	4
Charge calculation	(+3) + [6 × (0)] = +3	(+2) + [6 × (–1)] = –4	(0) + [4 × (0)] = 0
Final formula and charge	$\mathbf{[Co(NH_3)_6]^{3+}}$	$\mathbf{[Fe(CN)_6]^{4-}}$	$\mathbf{[Ni(CO)_4]}$

Q.33 [Short] Find the oxidation state of the metal in $[Fe(CN)_6]^{3-}$ and $[Cu(NH_3)_4]^{2+}$.

Charge on metal = charge on sphere – total charge of ligands. In $[Fe(CN)_6]^{3-}$: x + 6(–1) = –3, so x = +3. In $[Cu(NH_3)_4]^{2+}$: x + 4(0) = +2, so x = +2.

18.5 GEOMETRY OF TRANSITION METAL COMPLEXES

Q.34 [Long] Describe the common geometries of transition metal complexes with bond angles and examples.

The geometry (shape) of a transition metal complex depends on:

1. the **electronic configuration** of the metal;
2. the **coordination number** (number of donor atoms attached) of the metal;
3. the **nature of the ligands**.

Geometry	C.N.	Bond angle	Description	Examples
Linear	2	180°	Two ligands on opposite sides of the central metal ion	$[Ag(NH_3)_2]Cl$, $K[Ag(CN)_2]$
Square planar	4	90°	Metal ion and four ligands in the same plane, at the corners of a square; characteristic of d^8 ions (Ni^{2+}, Pt^{2+}) with ligands like NH_3 and CN^-	$[Ni(CN)_4]^{2-}$, $[Pt(NH_3)_4]^{2+}$
Tetrahedral	4	109.5°	Four ligands at the corners of a tetrahedron; common when the metal ion has a d^7, d^8 or d^{10} configuration	$[CoCl_4]^{2-}$, $[MnCl_4]^{2-}$
Octahedral	6	90° (180° for opposite ligands)	Six ligands along the x, y and z axes: four in a square plane, one above and one below it. The most common geometry	$[Cr(NH_3)_6]^{3+}$, $[Co(en)_3]^{3+}$

Q.35 [Short] Differentiate between square planar and tetrahedral complexes.

Square planar	Tetrahedral
Metal ion and four ligands lie in the same plane.	Four ligands are at the corners of a tetrahedron.
Bond angle 90°.	Bond angle 109.5°.
Characteristic of d^8 ions (Ni^{2+}, Pt^{2+}) with NH_3, CN^-.	Common for d^7, d^8 and d^{10} ions.
$[Ni(CN)_4]^{2-}$, $[Pt(NH_3)_4]^{2+}$	$[CoCl_4]^{2-}$, $[MnCl_4]^{2-}$

Quick Check 18.3

Q.36 [Short] **a) How Lewis acid base concept explains the transition metal complex formation?**

The central transition metal atom or ion is a Lewis acid: it accepts electron pairs into its vacant d-orbitals. Each ligand is a Lewis base: it donates a lone pair. This donation and acceptance forms coordinate covalent bonds and gives a stable complex.

Q.37 [Short] **b) Determine the overall charge of a complex ion formed from a Cr^{3+} metal ion with six neutral H_2O ligands and a coordination number of 6. Show your calculation.**

Charge = metal charge + (number of ligands × ligand charge) = (+3) + [6 × (0)] = **+3**. The complex ion is $[Cr(H_2O)_6]^{3+}$.

Q.38 [Short] **c) Predict the likely geometry for the complex ion $[MnCl_4]^{2-}$, given that the central metal ion has a d^5 electron configuration and a coordination number of 4.**

Tetrahedral, with bond angles of 109.5°. Square planar geometry is characteristic of d^8 ions, and the textbook lists $[MnCl_4]^{2-}$ as a tetrahedral complex.

Q.39 [Short] **d) EDTA is described as a hexadentate ligand. Explain what "hexadentate" means in terms of coordinate covalent bonds formed with a single metal ion.**

Hexadentate means that one EDTA ion forms **six** coordinate covalent bonds with a single metal ion, through six donor atoms (four oxygen atoms and two nitrogen atoms).

Q.40 [Short] **e) Identify the two essential features of a transition metal ion that enable it to form complex ions.**

(i) It has vacant (partially filled or empty) d-orbitals. (ii) These orbitals are energetically accessible, i.e., low enough in energy to accept the lone pairs donated by ligands.

18.6 COLOUR OF COMPLEXES

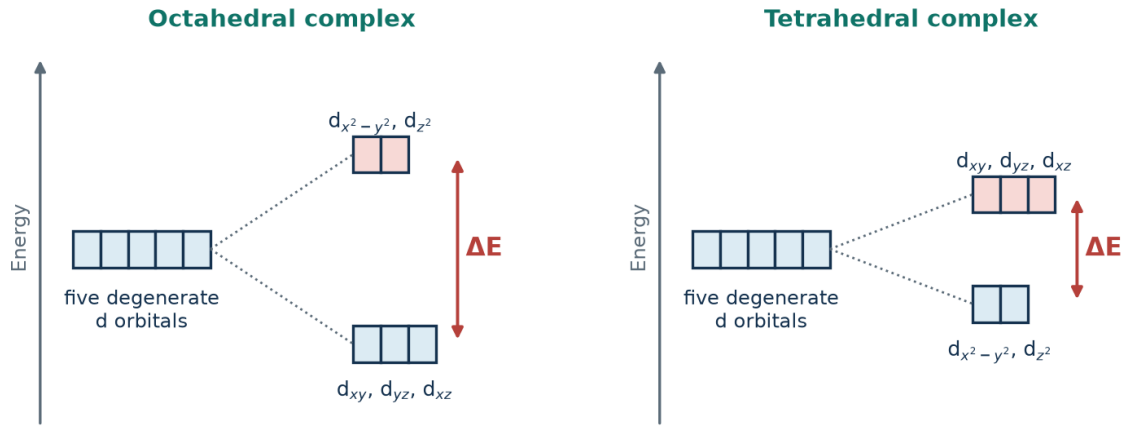
Q.41 [Short] **What are degenerate and non-degenerate d-orbitals?**

The d-orbitals have a characteristic cloverleaf shape, generally four lobes (two dumbbells in a plane). There are five of them: d_{xy}, d_{xz}, d_{yz}, $d_{x^2-y^2}$ and d_{z^2}. In an isolated atom all five have the same energy (**degenerate**). When ligands approach the metal ion to form a complex, they create an electric field that splits the five d-orbitals into two groups of different energy (**non-degenerate**). The energy difference between the two sets is represented by **Δ (ΔE)**.

Q.42 [Long] **Describe the splitting of d-orbitals in octahedral and tetrahedral complexes.**

Octahedral complexes (six ligands), e.g., $[Ti(H_2O)_6]^{3+}$: the d-orbitals split into three lower-energy orbitals (d_{xy}, d_{xz}, d_{yz}) and two higher-energy orbitals ($d_{x^2-y^2}$, d_{z^2}).

Tetrahedral complexes (four ligands), e.g., $[CuCl_4]^{2-}$: the splitting is inverted, with three higher-energy orbitals (d_{xy}, d_{xz}, d_{yz}) and two lower-energy orbitals ($d_{x^2-y^2}$, d_{z^2}). The magnitude of splitting is smaller than in octahedral complexes.



Splitting of the d-orbitals in octahedral and tetrahedral fields

Q.43 [Short] Describe the shapes of the d_{xy} and d_{z^2} orbitals.

The d_{xy} orbital has four lobes lying in the xy-plane, pointing *between* the x and y axes (cloverleaf shape). The d_{z^2} orbital has two lobes along the z-axis with a ring (collar) of electron density around the middle in the xy-plane.

The textbook mentions only the general cloverleaf shape; the details of these two orbitals are standard chemistry.

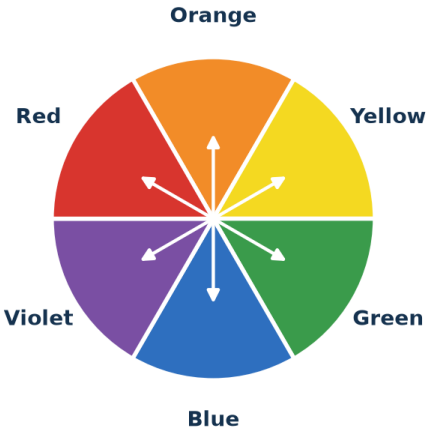
18.6.1 Colour Formation

Q.44 [Long] Explain why transition metal complexes are coloured.

When white light falls on a compound: if all of it is absorbed, the complex appears black; if all is reflected or transmitted, it appears white; if part is absorbed, the complex shows the colour of the remaining light. The colour we see is the **complementary colour** (opposite on the colour wheel) of the light absorbed. For example, a complex that absorbs yellow light appears violet.

Electrons in a lower-energy d-orbital absorb light and are promoted (excited) to a higher-energy d-orbital. This is called a **d-d transition**. The unabsorbed light is transmitted and decides the colour. The energy needed corresponds to a specific frequency (ν) and wavelength (λ) of visible light by Planck's equation:

$$\Delta E = h\nu = hc/\lambda \text{ (h = Planck's constant, c = speed of light)}$$



Colour wheel: opposite colours are complementary

Complementary pairs: Red ↔ Green, Orange ↔ Blue, Yellow ↔ Violet.

Q.45 [Short] When does a complex appear black, and when white?

If the white light falling on it is absorbed completely, the complex appears black. If the whole light is reflected or transmitted, it appears white.

Q.46 [Long] Describe the effect of different ligands on ΔE and on the colour of a complex.

The nature of the ligand affects the magnitude of the splitting, ΔE. Ligands like NH_3 and CN^- cause larger splitting; for ligands like Cl^- and I^-, ΔE is smaller. A larger ΔE means light of higher frequency (shorter wavelength), in the violet or blue region, is absorbed, so the compound reflects yellow light and looks yellow. Complexes that absorb low-energy light (red/orange) appear blue or green. Order of ligands by their ability to split the d-orbitals (high ΔE to low ΔE):

$$CN^- > en > NH_3 > H_2O > OH^- > Cl^- > Br^- > I^-$$

Q.47 [Short] Why are Zn^{2+} and Sc^{3+} compounds colourless?

Colour needs a d-d transition, which requires a partially filled d-subshell. Zn^{2+} ($3d^{10}$) has no vacant place in the higher d-orbitals and Sc^{3+} ($3d^0$) has no d electron to promote, so no visible light is absorbed.

This application is not stated in the textbook; it follows from its explanation of d-d transitions.

Quick Check 18.4

Q.48 [Short] a) How does the presence of ligands affect the degeneracy of d-orbitals in a complexion?

In an isolated atom the five d-orbitals are degenerate. The approaching ligands create an electric field that splits them into two sets of different energy, so they become non-degenerate. The energy gap between the sets is ΔE.

Q.49 [Short] **b) Describe the key difference in d-orbital splitting patterns between octahedral complexes and tetrahedral complexes.**

In octahedral complexes three orbitals (d_{xy}, d_{xz}, d_{yz}) are lower and two ($d_{x^2-y^2}$, d_{z^2}) are higher in energy. In tetrahedral complexes the pattern is inverted (three higher, two lower) and the splitting is smaller.

Q.50 [Short] **c) Explain the relationship between the absorbed light energy (ΔE) and the observed colour of a transition metal complex. Reference Planck's equation (ΔE=hν=hc/λ).**

The complex absorbs light whose energy equals ΔE; by $\Delta E = h\nu = hc/\lambda$ this fixes its frequency and wavelength. A larger ΔE means higher frequency (shorter wavelength) is absorbed. The remaining light is transmitted, so the observed colour is the complementary colour of the light absorbed.

Q.51 [Short] **d) If a complex containing the ligand Cl^- absorbs light in the red/orange region of the spectrum, what general colour range would you expect the complex to appear?**

It would appear **blue or green**, the complementary colours of orange and red.

Q.52 [Short] **e) Explain the phenomenon (d-d transitions) responsible for the colour of transition metal compounds.**

An electron in a lower-energy d-orbital absorbs light from the visible region and is promoted (excited) to a higher-energy d-orbital. This is a d-d transition. The unabsorbed light is transmitted and gives the compound its colour.

18.7 LIGAND EXCHANGE REACTIONS OF COMPLEXES

Q.53 [Short] **What is a ligand exchange (substitution) reaction? When does it occur?**

It is a reaction in which the ligands of a complex are replaced by other ligands. It occurs if the new complex to be formed is more stable. It changes the properties of the complex, such as colour, structure and reactivity.

18.7.1 Ligand Exchange Reactions of Copper(II) Complexes

Q.54 [Long] **Describe the ligand exchange reactions of the copper(II) ion with hydroxide ions, ammonia and chloride ions.**

$Cu^{2+}_{(aq)}$ is actually the complex ion $[Cu(H_2O)_6]^{2+}$, which gives the blue colour of copper(II) salt solutions such as copper sulphate. On adding sodium hydroxide, a light blue precipitate forms; two water ligands are replaced by two hydroxide ions.

$$[Cu(H_2O)_6]^{2+}_{(aq)} + 2OH^-_{(aq)} \rightarrow [Cu(H_2O)_4(OH)_2]_{(s)} + 2H_2O_{(l)}$$

blue solution → light blue precipitate

If concentrated ammonia solution is added, the light blue precipitate dissolves to give a **dark blue solution**. The water ligands of $[Cu(H_2O)_6]^{2+}$ can also be exchanged directly

with ammonia, four water molecules being replaced. For simplicity $[Cu(NH_3)_4(H_2O)_2]^{2+}$ is usually written $[Cu(NH_3)_4]^{2+}$.

$$[Cu(H_2O)_6]^{2+}_{(aq)} + 4NH_{3(aq)} \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+}_{(aq)} + 4H_2O_{(l)}$$

On adding concentrated hydrochloric acid drop by drop, chloride ions replace the water ligands. A **yellow (yellow-green) solution** containing the tetrahedral complex ion $[CuCl_4]^{2-}$ is obtained.

$$[Cu(H_2O)_6]^{2+}_{(aq)} + 4Cl^-_{(aq)} \rightarrow [CuCl_4]^{2-} + 6H_2O$$
blue solution → yellow solution

18.7.2 Ligand Exchange Reactions of Cobalt(II) Complexes

Q.55 [Long] **Describe the ligand exchange reactions of the cobalt(II) ion with hydroxide ions, ammonia and chloride ions.**

$Co^{2+}_{(aq)}$ is the complex ion $[Co(H_2O)_6]^{2+}$, which gives the pink colour of aqueous cobalt(II) sulphate. On adding sodium hydroxide solution, a **blue precipitate** of cobalt(II) hydroxide forms, which may turn pink on standing.

$$[Co(H_2O)_6]^{2+}_{(aq)} + 2OH^-_{(aq)} \rightarrow [Co(H_2O)_4(OH)_2]_{(s)} + 2H_2O_{(l)}$$
pink solution → blue precipitate

When excess ammonia solution is added to the cobalt(II) hydroxide precipitate, ammonia ligands replace the water and hydroxide ligands, forming the soluble ammine complex, hexaamminecobalt(II) ion. The solution turns **yellow**. Water ligands in $[Co(H_2O)_6]^{2+}$ can also be exchanged directly with concentrated ammonia.

$$[Co(H_2O)_4(OH)_2]_{(s)} + 6NH_{3(aq)} \rightarrow [Co(NH_3)_6]^{2+}_{(aq)} + 4H_2O_{(l)} + 2OH^-_{(aq)}$$
blue precipitate → yellow solution

$$[Co(H_2O)_6]^{2+} + 6NH_{3(aq)} \rightarrow [Co(NH_3)_6]^{2+}_{(aq)} + 6H_2O_{(l)}$$

Adding concentrated hydrochloric acid drop by drop to aqueous cobalt(II) ions gives a **blue solution** containing the tetrahedral complex $[CoCl_4]^{2-}$. Aqueous cobalt(II) ions usually form tetrahedral complexes with monodentate anionic ligands such as Cl^-, OH^- and SCN^-.

$$[Co(H_2O)_6]^{2+}_{(aq)} + 4Cl^-_{(aq)} \rightarrow [CoCl_4]^{2-}_{(aq)} + 6H_2O_{(l)}$$
pink solution → blue solution

18.7.3 Effect of Ligand Exchange on the Colour of Complexes

Q.56 [Short] **Why does the colour of a complex change when its ligands are exchanged?**

The exchange of ligands changes the magnitude of splitting (ΔE) for the metal ion. So the frequency of light absorbed also changes, and the observed colour changes. In each case the new ligands (NH_3 or Cl^-) make a more stable complex than the original H_2O ligands, with a different ΔE, a different absorbed frequency and a different observed colour.

Metal ion	Initial complex (colour)	Ligand exchange reaction	Final complex (colour)
Cu^{2+}	$[Cu(H_2O)_6]^{2+}$ (blue)	Add conc. NH_3 solution	$[Cu(NH_3)_4(H_2O)_2]^{2+}$ (deep blue)
Cu^{2+}	$[Cu(H_2O)_6]^{2+}$ (blue)	Add conc. HCl solution	$[CuCl_4]^{2-}$ (yellow/green)
Co^{2+}	$[Co(H_2O)_6]^{2+}$ (pink)	Add conc. NH_3 solution	$[Co(NH_3)_6]^{2+}$ (yellow)
Co^{2+}	$[Co(H_2O)_6]^{2+}$ (pink)	Add conc. HCl solution	$[CoCl_4]^{2-}$ (blue)

Quick Check 18.5

Q.57 [Short] **a) Write the equation for the reaction, where $[Cu(H_2O)_4(OH)_2]_{(s)}$ dissolves in excess ammonia.**

$$[Cu(H_2O)_4(OH)_2]_{(s)} + 4NH_{3(aq)} \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+}{}_{(aq)} + 2H_2O_{(l)} + 2OH^-_{(aq)}$$

The light blue precipitate dissolves to give a dark blue solution.

Q.58 [Short] **b) Pink $[Co(H_2O)_6]^{2+}$ turns blue with conc. HCl. Explain this colour change using d-orbital splitting.**

The Cl^- ligands replace the water ligands and the complex changes from octahedral $[Co(H_2O)_6]^{2+}$ to tetrahedral $[CoCl_4]^{2-}$. Cl^- causes less splitting than H_2O, and tetrahedral splitting is smaller than octahedral, so ΔE decreases. Light of lower frequency is now absorbed, and the complementary colour seen changes from pink to blue.

Q.59 [Short] **c) For the reaction: $[Co(H_2O)_6]^{2+} + 4Cl^- \rightarrow [CoCl_4]^{2-} + 6H_2O$, identify the coordination number and geometry of the product complex.**

In $[CoCl_4]^{2-}$ the coordination number is **4** and the geometry is **tetrahedral**.

18.8 STABILITY OF COMPLEXES AND STABILITY CONSTANT, Kstab

Q.60 [Long] **Define the stability constant. Write the K_{stab} expression for a complex and explain its significance.**

The **stability constant (K_{stab})** is the equilibrium constant (K_c) for the formation of a complex ion in a solvent from its constituent ions or molecules. In aqueous solution water is the initial ligand; a new ligand competes with water to bind the metal ion.

$$[Cu(H_2O)_6]^{2+}_{(aq)} + 4Cl^-_{(aq)} \rightleftharpoons [CuCl_4]^{2-}_{(aq)} + 6H_2O_{(l)}$$
$$K_{stab} = [[CuCl_4]^{2-}] / ([[Cu(H_2O)_6]^{2+}] [Cl^-]^4)$$

The concentration of water, $[H_2O]$, is omitted because water is the solvent and its concentration remains constant. Stability constants are usually given on a $\log_{10}$ scale, which makes very large numbers easier to compare. The higher the $\log_{10}K_{stab}$, the more stable the complex, and the equilibrium lies far to the right. The lower it is, the less stable the complex, and its ligands can be easily displaced.

Complex	$\log_{10} K_{stab}$
$[CuCl_4]^{2-}$	5.6
$[Cu(NH_3)_6]^{2-}$ (as printed)	13.1
$[Cu(EDTA)]^{2-}$	18.8

The second complex is printed in Table 18.5 with the charge 2–; an ammine complex of copper(II) carries the charge 2+.

Q.61 [Short] What is the chelating effect?

Complexes with polydentate ligands have higher stability constants than those with monodentate ligands. This is called the **chelating effect** of the ligands, e.g. $[Cu(H_2O)_6]^{2+} + 3en \rightarrow [Cu(en)_3]^{2+} + 6H_2O$.

Q.62 [Short] Explain ligand exchange in terms of K_{stab} values.

A ligand with a higher K_{stab} can displace a ligand with a lower K_{stab}. For example, adding ammonia to a chloride complex turns the solution blue, because the ammonia complex is significantly more stable.

$$[CuCl_4]^{2-}_{(aq)} + 6NH_{3(aq)} \rightarrow [Cu(NH_3)_6]^{2+} + 4Cl^-_{(aq)}$$

Q.63 [Short] What is the effect of dilution on a stable complex?

Yes. If a stable complex is diluted with a massive amount of water, the equilibrium can be forced back to the left, re-forming the original hydrated ion. Water is itself a ligand; a very large excess of it competes with the other ligand and, by Le Chatelier's principle, the position of equilibrium shifts towards the aqua complex.

Quick Check 18.6

Q.64 [Short] a) Write the K_{stab} equilibrium expression for the formation of $[Cu(NH_3)_4]^{2+}$ from $[Cu(H_2O)_6]^{2+}$ and NH_3.

$$[Cu(H_2O)_6]^{2+} + 4NH_3 \rightleftharpoons [Cu(NH_3)_4]^{2+} + 6H_2O$$
$$K_{stab} = [[Cu(NH_3)_4]^{2+}] / ([[Cu(H_2O)_6]^{2+}] [NH_3]^4)$$

$[H_2O]$ is not included.

Q.65 [Short] **b) Explain why polydentate ligands form more stable complexes using K_{stab} data ($\log_{10}K_{stab}$) data for $[CuCl_4]^{2-}$ (5.6) and $[Cu(EDTA)]^{2-}$ (18.8).**

$\log_{10}K_{stab}$ of $[Cu(EDTA)]^{2-}$ (18.8) is far higher than that of $[CuCl_4]^{2-}$ (5.6), so its equilibrium lies much further to the right. Cl^- is monodentate, while one EDTA ion holds the metal through six dative bonds. This extra stability of complexes with polydentate ligands is the chelating effect.

Q.66 [Short] **c) Write the balanced equation for the first step of the stepwise substitution of water ligands by ammonia in $[Cu(H_2O)_6]^{2+}$.**

$$[Cu(H_2O)_6]^{2+}_{(aq)} + NH_{3(aq)} \rightleftharpoons [Cu(NH_3)(H_2O)_5]^{2+}_{(aq)} + H_2O_{(l)}$$

18.9 STEREOISOMERISM IN TRANSITION METAL COMPLEXES

Q.67 [Short] **What are stereoisomers? Name the types found in complexes.**

Compounds that have the same chemical formula and bonds but differ in the arrangement of the ligands in space are called **stereoisomers**. There are two types: **geometrical (cis-trans) isomerism** and **optical isomerism**.

18.9.1 Cis-Trans Isomerism

Q.68 [Long] **Explain cis-trans isomerism in square planar and octahedral complexes.**

In cis-trans isomers, identical ligands are present at either adjacent positions (90° angle, **cis**) or opposite positions (180° angle, **trans**). It is common in square planar complexes (e.g., $[Pt(NH_3)_2Cl_2]$) and octahedral complexes (e.g., $[Co(NH_3)_4Cl_2]$).

In *cis*-$[Pt(NH_3)_2Cl_2]$ (cisplatin) the two chlorine atoms are adjacent, at 90° in the square plane. In *trans*-$[Pt(NH_3)_2Cl_2]$ (transplatin) they are opposite, at 180°.



Geometrical isomers of $[Pt(NH_3)_2Cl_2]$

The properties of the two isomers are different. Cisplatin is used as an anti-cancer drug; transplatin does not show this property. In transplatin the bond dipoles of the identical ligands point in opposite directions and cancel, so the molecule is **non-polar**. In cisplatin they do not cancel, giving a non-zero dipole moment, so it is **polar**.

$[Co(NH_3)_4(H_2O)_2]^{3+}$ is an octahedral complex of the type $[MA_4B_2]$. In the cis isomer the two H_2O ligands occupy adjacent positions at 90° to each other; in the trans isomer they occupy opposite positions at 180°.

Q.69 [Short] Why is cisplatin polar while transplatin is non-polar?

In transplatin the bond dipoles of the identical ligands point in opposite directions and cancel each other. In cisplatin the similar bond moments do not cancel, so there is a net dipole moment.

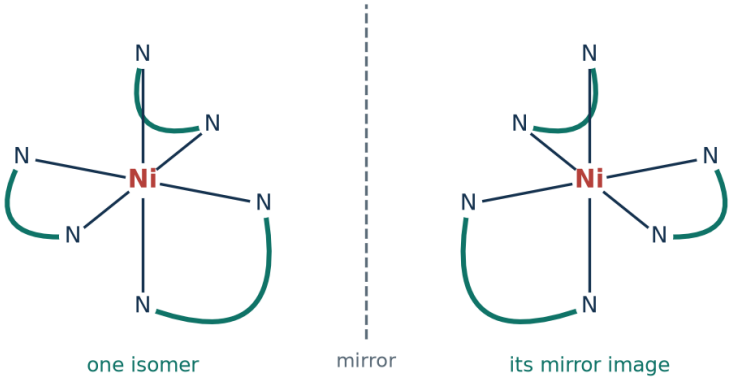
Q.70 [Short] Write a note on cisplatin as an anticancer drug.

Cisplatin, *cis*-diamminedichloroplatinum(II), is a landmark inorganic anticancer drug. Its discovery was accidental: platinum electrodes were found to inhibit bacterial cell division. It targets rapidly dividing cancer cells by entering the cell and undergoing ligand exchange, in which water replaces its chloride ions. The “activated” drug then binds to guanine bases in DNA, creating cross-links that link the double helix. This structural damage prevents DNA replication and triggers programmed cell death (apoptosis).

18.9.2 Optical Isomerism

Q.71 [Long] Explain optical isomerism in transition metal complexes.

Optical isomers are non-superimposable mirror images of each other. Optical isomerism is commonly shown by octahedral complexes with bidentate ligands, e.g., the nickel(II) complex with 1,2-diaminoethane, $[Ni(en)_3]^{2+}$. Its two isomers are mirror images that cannot be superimposed; such pairs are called **enantiomers**. They rotate the plane of polarized light in opposite directions: the isomer that rotates it to the right is **dextrorotatory (+)**, the one that rotates it to the left is **laevorotatory (–)**.



The two non-superimposable optical isomers of $[Ni(en)_3]^{2+}$ (each curve is one en ligand)

Q.72 [Short] Differentiate between dextrorotatory and laevorotatory isomers.

Dextrorotatory	Laevorotatory
Rotates the plane of polarized light in the right-hand direction.	Rotates the plane of polarized light in the left-hand direction.
Shown by the sign “+”.	Shown by the sign “–”.

Quick Check 18.7

Q.73 [Short] **a) Which type of geometries shows cis-trans isomerism?**

Square planar and octahedral geometries.

Q.74 [Short] **b) What type of ligand is most often required for a complex to show optical isomerism?**

A bidentate ligand (in an octahedral complex), e.g., 1,2-diaminoethane.

Q.75 [Short] **c) For the complex $[Ni(en)_3]^{2+}$, state the coordination number and the type of ligand.**

Coordination number = **6** (three *en* ligands, each forming two dative bonds). *en* (1,2-diaminoethane) is a neutral **bidentate** ligand.

Q.76 [Short] **d) Identify the type of isomerism displayed by the complex $[Ni(NH_2CH_2CH_2NH_2)_3]^{2+}$, and describe how the two different isomers interact with plane-polarized light.**

It shows **optical isomerism**. The two isomers rotate the plane of polarized light in opposite directions: one to the right (dextrorotatory, +) and the other to the left (laevorotatory, –).

Q.77 [Short] **e) Define optical isomers (enantiomers) in terms of their mirror images and superimposability.**

Optical isomers (enantiomers) are a pair of molecules that are mirror images of each other and cannot be superimposed on each other.

18.10 REDOX REACTIONS OF TRANSITION METALS

Q.78 [Short] **How is the feasibility of a redox reaction predicted from E° values?**

Transition elements can exist in multiple oxidation states. The feasibility of a redox reaction is found by comparing the standard electrode potentials (E°) of the half-reactions. A reaction is feasible if the overall cell potential is **positive**.

$$E°_{cell} = E°_{reduction} - E°_{oxidation}$$

The species with the more positive E° value is reduced (gains electrons); the species with the more negative (less positive) value is oxidized (loses electrons).

18.10.1 Reaction of Acidified Manganate(VII) Ions with Ethanedioate Ions

Q.79 [Long] **Analyse the reaction of acidified manganate(VII) ions with ethanedioate ions.**

The half-reaction with the more positive E° is more likely to undergo reduction. So MnO_4^- is reduced to Mn^{2+} (it is the oxidizing agent) and $C_2O_4^{2-}$ is oxidized to CO_2 (it is the reducing agent).

$$2CO_{2(g)} + 2e^- \rightarrow C_2O_4^{2-}{}_{(aq)} \quad E° = +0.49\ V$$
$$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O_{(l)} \quad E° = +1.52\ V$$

The electrons lost must equal the electrons gained, so the reduction half-equation is multiplied by 2 and the oxidation half-equation by 5:

$$2 \times [MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O_{(l)}]$$
$$5 \times [C_2O_4^{2-}{}_{(aq)} \rightarrow 2CO_{2(g)} + 2e^-]$$
$$2MnO_4^- + 5C_2O_4^{2-}{}_{(aq)} + 16H^+ \rightarrow 2Mn^{2+} + 10CO_{2(g)} + 8H_2O_{(l)}$$
$$E°_{cell} = (+1.52) - (+0.49) = +1.03\ V$$

The large positive $E°_{cell}$ (+1.03 V) shows that the forward reaction is feasible.

The textbook prints the overall equation as "$2MnO_4^- + 5C_2O_4^{2-} + 8H^+ \rightarrow 5Mn^{2+} + 2CO_2 + 8H_2O$", which is not balanced. The balanced equation is given above; $E°_{cell} = +1.03$ V is as in the textbook.

18.10.2 Reaction of Manganate(VII) Ion with Iron(II) Ion

Q.80 [Long] **Analyse the reaction of manganate(VII) ions with iron(II) ions in acid solution.**

MnO_4^- is reduced and Fe^{2+} is oxidized, as indicated by the potential values.

$$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O_{(l)} \quad E° = +1.52\ V$$
$$Fe^{3+} + e^- \rightarrow Fe^{2+} \quad E° = +0.77\ V$$
$$5 \times [Fe^{2+} \rightarrow Fe^{3+} + e^-]$$
$$MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O_{(l)}$$
$$E°_{cell} = (+1.52) - (+0.77) = +0.75\ V$$

The positive value shows that the reaction proceeds to the right, i.e., it is feasible.

The textbook adds the two values and prints $E° = +2.29$ V (and writes the product as $5Fe^{2+}$). Using the textbook's own formula $E°_{cell} = E°_{reduction} - E°_{oxidation}$, the value is $+1.52 - 0.77 = +0.75$ V. Either way the value is positive and the reaction is feasible.

18.10.3 Reaction Between Copper(II) Ion and Iodide Ion

Q.81 [Long] **Analyse the reaction between copper(II) ions and iodide ions. Why does it proceed although E° is negative?**

When an excess of potassium iodide is added to aqueous copper(II) sulphate, a white precipitate of copper(I) iodide forms in a red-brown solution of iodine:

$$2Cu^{2+}{}_{(aq)} + 4I^-{}_{(aq)} \rightarrow 2CuI_{(s)} + I_{2(aq)}$$
$$\tfrac{1}{2}I_{2(aq)} + e^- \rightarrow I^-{}_{(aq)} \quad E° = +0.54\ V$$
$$Cu^{2+}{}_{(aq)} + e^- \rightarrow Cu^+{}_{(aq)} \quad E° = +0.15\ V$$

The I_2/I^- couple has the higher E°, so it would be more likely to undergo reduction; the Cu^{2+}/Cu^+ couple has the lower E°, suggesting Cu^+ is a better reducing agent than I^-. So

the calculated net standard potential is **negative** (0.15 – 0.54 = –0.39 V) and the reaction looks not feasible.

However, the reaction does proceed forward. CuI is continuously precipitated out of the solution, which removes the Cu^{+} product and shifts the equilibrium to the right (**Le Chatelier's principle**). The amount of I_2 formed can be found by titrating with a thiosulphate solution.

Q.82 [Short] **How can the amount of iodine formed in this reaction be measured?**

By titrating the iodine formed with a thiosulphate solution.

Quick Check 18.8

Q.83 [Short] **a) State the principle used to determine the feasibility of a redox reaction using standard electrode potentials (E°), as described in the text.**

The E° values of the two half-reactions are compared. A reaction is feasible if the overall cell potential, $E°_{cell} = E°_{reduction} – E°_{oxidation}$, is positive. The species with the more positive E° is reduced; the one with the less positive E° is oxidized.

Q.84 [Short] **b) For the reaction between MnO_4^- and Fe^{2+}, the relevant reduction potentials are MnO_4^-/Mn^{2+} ($E° = +1.52$ V) and Fe^{3+}/Fe^{2+} ($E° = +0.77$ V). Identify the oxidizing agent and the reducing agent in this specific feasible reaction.**

MnO_4^- has the more positive E°, so it is reduced: it is the **oxidizing agent**. Fe^{2+} is oxidized to Fe^{3+}: it is the **reducing agent**.

Q.85 [Short] **c) Explain why water is omitted when calculating the $E°_{cell}$ for the reaction between MnO_4^- and $C_2O_4^{2-}$ ions, even though it appears in the overall balanced equation.**

$E°_{cell}$ is calculated only from the E° values of the two half-reactions (+1.52 V and +0.49 V). Water is the solvent: it is neither oxidized nor reduced and has no electrode potential of its own in this reaction, so it does not enter the calculation.

The textbook does not state this reason directly; the answer follows from the method it uses.

studysapiens.com